



SHARIF UNIVERSITY OF TECHNOLOGY

Department of Mechanical Engineering

Fluid Mechanics I Project Report

Numerical Investigation of Hemodynamic Flow and Pressure Drop in Stenosed Carotid Arteries

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Abstract

Carotid artery stenosis, driven by atherosclerotic plaque deposition, remains one of the primary precursors to ischemic stroke and severe cerebrovascular disorders. In this study, a computational fluid dynamics (CFD) investigation is conducted to analyze steady, laminar, incompressible hemodynamic flow through stenosed arterial segments. Two geometrical constriction severities—mild (30% diameter reduction) and moderate (40% diameter reduction)—are modeled at an axial location of $x/L = 0.5$ in an artery of diameter $D = 3.5$ mm and length $L = 61.6$ mm. A pressure-based finite-volume solver employing the SIMPLE algorithm is utilized. Comprehensive grid independence studies are performed across coarse, medium, and fine grids to ensure numerical verification with relative discretization errors below 2%. The spatial distributions of velocity, static pressure, dynamic pressure, and total pressure are analyzed alongside cross-sectional velocity profiles at the stenosis throat. The pressure loss coefficient (ξ) is quantified as 216.07 for the 30% stenosis and 234.47 for the 40% stenosis, demonstrating that a 10% increment in radial occlusion induces an 8.5% rise in hydraulic head loss. Furthermore, a parametric rheological study substituting blood with glycerin ($\mu = 0.799$ Pa·s, $\rho = 1259.9$ kg/m³) is conducted; results show an over eighteen-fold increase in static pressure drop ($\Delta p = 14,703$ Pa vs. 798 Pa) and a loss coefficient of $\xi = 3646.5$, proving that at low Reynolds numbers, viscous dissipation dominates over inertial effects.

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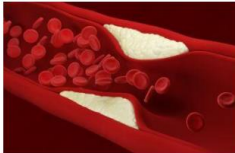
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1 Introduction and Scientific Background

Carotid artery stenosis is a pathological narrowing of the carotid bifurcation or internal carotid artery lumen, caused predominantly by the progressive deposition of lipid-rich plaques, fibrous tissue, and cellular debris beneath the endothelial intima (atherosclerosis). The carotid arteries serve as the principal vascular conduit supplying oxygenated blood to the brain and cerebral hemispheres.

When plaque builds up, it leads to luminal constriction, which locally constricts the blood flow path. From a hemodynamic standpoint, this constriction creates localized flow acceleration, sharp trans-stenotic pressure gradients, elevated wall shear stress (WSS) at the throat, and flow separation or stagnation zones downstream of the narrowing. These hemodynamic anomalies are of paramount clinical significance:

1. **Thrombus Formation and Embolism:** High shear stresses at the constriction throat can activate blood platelets and cause endothelial denudation. Concurrently, downstream recirculating and low-shear regions favor platelet aggregation and thrombus formation, which may dislodge and travel to distal cerebral vessels, precipitating transient ischemic attacks (TIAs) or full-blown strokes.
2. **Elevated Cardiac Afterload and Perfusion Deficit:** Severe narrowing imposes substantial hydraulic resistance, requiring higher upstream perfusion pressures to maintain adequate cerebral blood supply.



- Stenosis of Carotid Artery

(a) Atherosclerotic plaque accumulation obstructing lumen

(b) Comparison of dilated, normal, and constricted vessels

Figure 1: Physiological and pathological representations of arterial stenosis.

Computational fluid dynamics (CFD) provides a non-invasive, high-resolution modality for quantifying patient-specific arterial flow fields, trans-stenotic pressure gradients, and hydraulic head losses. The primary objective of the present investigation is to model the hemodynamic flow in an arterial segment with symmetric stenosis, evaluate the influence of stenosis severity on hydraulic performance, verify numerical accuracy via grid convergence, and investigate the sensitivity of pressure loss to fluid viscosity.

2 Governing Equations and Numerical Methodology

2.1 Flow Physics and Assumptions

The blood flow inside the medium-to-large arterial conduit is modeled under the following fundamental physical assumptions:

- **Steady-State Flow:** Time-averaged physiological flow conditions are assumed ($\partial/\partial t = 0$).
- **Incompressibility:** Under physiological velocities, blood density exhibits negligible divergence ($\nabla \cdot \vec{u} = 0$).
- **Laminar Regime:** For an arterial diameter $D = 3.5$ mm, mean blood velocity $u_{\text{in}} = 0.08$ m/s, density $\rho = 1063$ kg/m³, and dynamic viscosity $\mu = 0.043$ Pa · s, the Reynolds number is:

$$\text{Re} = \frac{\rho u_{\text{in}} D}{\mu} = \frac{1063 \times 0.08 \times 0.0035}{0.043} \approx 6.92 \quad (1)$$

Even at the throat of the 40% stenosis, where local velocities reach 0.43 m/s, the throat Reynolds number remains $\text{Re}_{\text{throat}} \approx 22$, which is well below the critical laminar-to-turbulent transition threshold ($\text{Re}_{\text{crit}} \approx 2000$).

- **Newtonian Rheology:** At characteristic shear rates encountered in conduit arteries ($\dot{\gamma} > 100$ s⁻¹), whole blood asymptotically exhibits constant Newtonian viscosity.

2.2 Governing Navier–Stokes Equations

Under the above assumptions, the conservation of mass and momentum in differential form are given by:

$$\nabla \cdot \vec{u} = 0 \quad (2)$$

$$\rho(\vec{u} \cdot \nabla) \vec{u} = -\nabla p + \mu \nabla^2 \vec{u} \quad (3)$$

where $\vec{u}$ represents the velocity vector, p is the static pressure, ρ is the fluid density, and μ is the dynamic viscosity.

2.3 Solver Configuration

Numerical simulations were carried out using ANSYS Fluent finite-volume solver:

- **Solver Type:** Pressure-based, double precision.
- **Pressure-Velocity Coupling:** Semi-Implicit Method for Pressure-Linked Equations (SIMPLE).
- **Spatial Discretization:** Second-Order Upwind scheme for the momentum equations; Standard scheme for pressure interpolation.
- **Convergence Criteria:** Scaled residuals for continuity, x -momentum, and y -momentum restricted to strictly less than 10^{-4} ($< 10^{-6}$ achieved for momentum).

2.4 Fluid Properties and Boundary Conditions

The physical properties of the fluids and prescribed boundary conditions are summarized in Table 1.

Table 1: Working fluid thermophysical properties and boundary conditions.

Parameter	Symbol / Unit	Blood (Primary)	Glycerin (Parametric)
Density	ρ [kg/m ³]	1063	1259.9
Dynamic Viscosity	μ [Pa · s]	0.043	0.799
Inlet Velocity	u_{in} [cm/s] ([m/s])	8.0 (0.08)	8.0 (0.08)
Outlet Static Pressure	p_{out} [Pa] ([mmHg])	11,999 (90.0)	11,999 (90.0)
Wall Boundary	—	Rigid, No-slip	Rigid, No-slip

The outlet pressure of 11,999 Pa (≈ 90 mmHg) corresponds directly to physiological human diastolic pressure.

2.5 Hydraulic Pressure Loss Coefficient (ξ)

To quantify the total hydrodynamic resistance across the stenosed vascular conduit, the non-dimensional pressure loss coefficient (ξ) is formulated as:

$$\xi = \frac{p_{\text{in}} - p_{\text{out}}}{\frac{1}{2}\rho u_{\text{in}}^2} = \frac{\Delta p}{q_{\text{in}}} \quad (4)$$

where p_{in} is the area-weighted average static pressure at the inlet, p_{out} is the static pressure at the outlet, and $q_{\text{in}} = \frac{1}{2}\rho u_{\text{in}}^2$ is the dynamic pressure at the inlet. For blood at $u_{\text{in}} = 0.08$ m/s:

$$q_{\text{in, blood}} = \frac{1}{2}(1063 \text{ kg/m}^3)(0.08 \text{ m/s})^2 = 3.4016 \text{ Pa} \quad (5)$$

For glycerin at $u_{\text{in}} = 0.08$ m/s:

$$q_{\text{in, gly}} = \frac{1}{2}(1259.9 \text{ kg/m}^3)(0.08 \text{ m/s})^2 = 4.0317 \text{ Pa} \quad (6)$$

3 Geometric Modeling and Computational Domain

The idealized arterial segment is modeled as an axisymmetric conduit with a smooth cosine-type symmetric stenosis. The principal dimensions are:

- Vessel unobstructed diameter: $D = 3.5$ mm ($R_0 = 1.75$ mm)
- Total vessel length: $L = 61.6$ mm ($L/D \approx 17.6$, ensuring sufficient entrance development length and post-stenotic recovery)
- Stenosis throat length: $L_{\text{stenosis}} = 6.0$ mm
- Axial stenosis location: Centered at $x/L = 0.5$ ($x = 30.8$ mm from inlet)

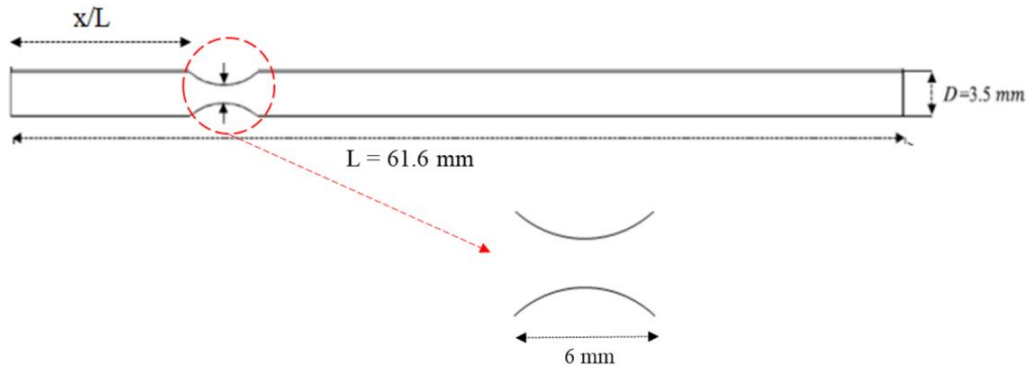


Figure 2: Schematic of the arterial geometry showing diameter $D = 3.5$ mm, total length $L = 61.6$ mm, stenosis location x/L , and the zoomed throat constriction region of length 6 mm.

The reduction in vessel opening (RVO), or occlusion percentage, defines the radial extent of luminal narrowing relative to the unconstricted lumen:

$$\text{RVO} = \left(1 - \frac{D_{\text{throat}}}{D}\right) \times 100\% \quad (7)$$

Two specific cases are examined in this study:

1. **Small Occlusion (30% Stenosis):** Throat diameter $D_{\text{throat}} = 0.70 \times 3.5 \text{ mm} = 2.45 \text{ mm}$ (carrying blood through 70% of original diameter).
2. **Large Occlusion (40% Stenosis):** Throat diameter $D_{\text{throat}} = 0.60 \times 3.5 \text{ mm} = 2.10 \text{ mm}$ (carrying blood through 60% of original diameter).

(a) Computational fluid domain for 30% stenosis at $x/L = 0.5$ (b) Computational fluid domain for 40% stenosis at $x/L = 0.5$

Figure 3: Verified geometric models for the 30% and 40% stenosed arterial segments.

4 Mesh Generation and Quality Verification

Spatial discretization was executed using structured and refined quadrilateral finite-volume cells. Boundary layer cells were applied near the vessel walls to accurately resolve velocity gradients and wall shear rates. To verify that numerical diffusion does not contaminate the solution, rigorous mesh quality metrics were evaluated.

4.1 Quality Criteria: Skewness and Orthogonality

Equilateral-volume skewness and orthogonal quality are standard measures of grid fidelity:

- **Skewness:** Ranges from 0 (equilateral, optimal) to 1 (degenerate). Values below 0.25 indicate excellent quality.
- **Orthogonal Quality:** Ranges from 0 (poor) to 1 (perfect perpendicular alignment with cell face normals). Values above 0.90 are classified as high quality.

Table 2 lists the mesh metrics for the medium grids of both geometric configurations.

Table 2: Detailed mesh quality metrics for the baseline medium grids.

Metric	30% Stenosis (Medium)		40% Stenosis (Medium)	
	Skewness	Orthogonal Quality	Skewness	Orthogonal Quality
Minimum	2.1958×10^{-7}	0.96195	1.0224×10^{-8}	0.92254
Maximum	0.24163	1.00000	0.25414	1.00000
Average	0.00767	0.99949	0.00980	0.99870
Std. Dev.	0.01921	0.00200	0.03148	0.00614
Nodes	31,062		33,015	
Elements	30,005		31,920	

Both grids exhibit mean orthogonal qualities exceeding 0.998 and average skewness below 0.010, ensuring numerical stability and minimal truncation error.

4.2 Grid Densities for Sensitivity Analysis

To substantiate grid independence, three levels of spatial resolution (coarse, medium, and fine) were synthesized for both cases, as tabulated in Table 3.

Table 3: Node and element counts across coarse, medium, and fine mesh levels.

Mesh Level	30% Stenosis		40% Stenosis	
	Nodes	Elements	Nodes	Elements
Coarse	11,654	11,023	11,738	11,103
Medium	31,062	30,005	33,015	31,920
Fine	113,202	111,313	121,466	119,356

5 Grid Independence Study and Numerical Convergence

5.1 Grid Convergence for Blood Flow (40% Stenosis)

To verify that the computed results are independent of cell dimensions, the area-weighted static pressure at the inlet (p_{in}) and the resultant pressure loss coefficient (ξ) were monitored across the three grid levels for the 40% stenosis case. Table 4 presents the computed parameters and the relative error between consecutive grids.

Table 4: Grid independence analysis for blood flow in the 40% stenosed artery.

Grid Level	Elements	p_{in} [Pa]	Δp [Pa]	ξ	Relative Error (%)
Coarse	11,103	12,815.08	816.08	239.91	—
Medium	31,920	12,796.60	797.60	234.47	2.26%
Fine	119,356	12,811.89	812.89	238.97	1.88%

The relative difference in the loss coefficient between the medium and fine meshes is only 1.88%, well below the standard 5% engineering tolerance. Consequently, the medium mesh with 31,920 elements achieves grid independence and was selected for subsequent parametric investigations, providing high fidelity without unnecessary computational overhead.

5.2 Iterative Residual Convergence

The scaled residuals for continuity, x -momentum, and y -momentum were monitored throughout the iterative solution procedure. As illustrated in Figure 4, monotonic exponential decay was achieved across all grid resolutions, with momentum residuals converging past 10^{-6} and continuity residuals settling well below 10^{-4} .

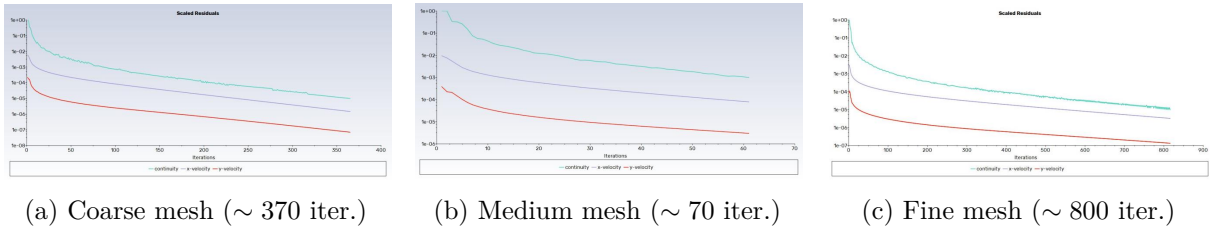
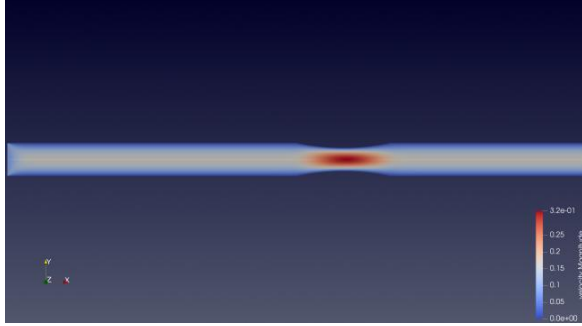


Figure 4: Convergence history of scaled residuals for blood flow across coarse, medium, and fine grids.

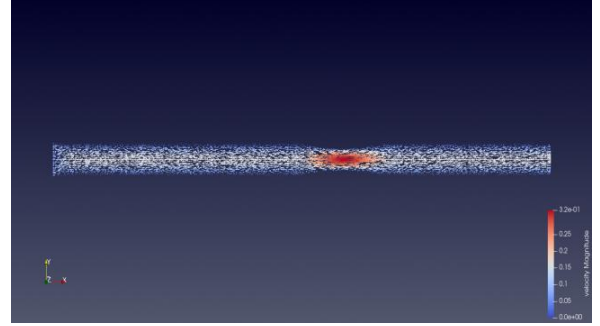
6 Hemodynamic Results and Hydraulic Performance Analysis

6.1 Velocity Fields and Streamline Patterns

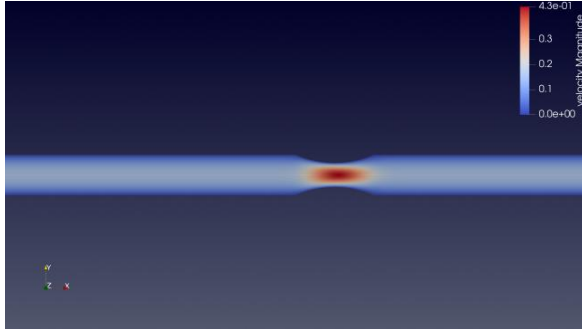
Figure 5 presents the velocity magnitude contours and corresponding flow streamlines for both the 30% and 40% stenosed arteries.



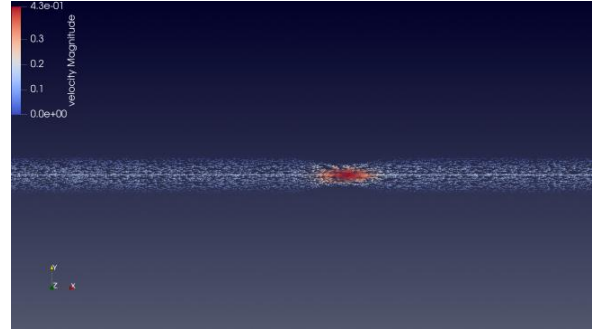
(a) Velocity contour (30%): $u_{\max} \approx 0.32$ m/s



(b) Streamlines: 30% stenosis



(c) Velocity contour (40%): $u_{\max} \approx 0.43$ m/s



(d) Streamlines: 40% stenosis

Figure 5: Velocity magnitude contours and streamline trajectories for 30% and 40% stenosis.

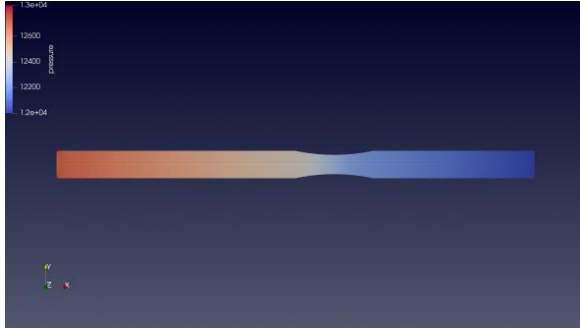
In accordance with the principle of mass conservation ($\dot{m} = \rho Au = \text{const}$), luminal narrowing requires a localized acceleration of blood flow:

- In the 30% stenosis, the peak throat velocity reaches 0.32 m/s, a 4-fold amplification compared to the inlet velocity (0.08 m/s).
- In the 40% stenosis, the peak throat velocity escalates to 0.43 m/s, representing a 5.4-fold amplification over the inlet velocity.

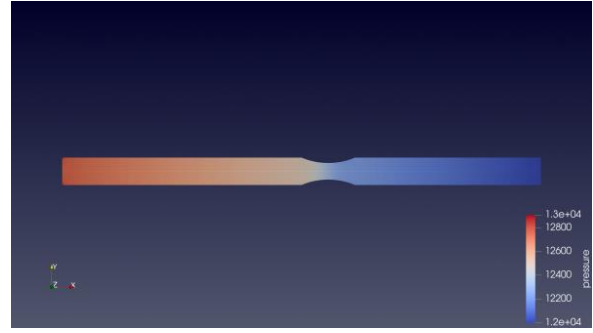
The streamline patterns confirm smooth acceleration toward the throat. Due to the low Reynolds number ($\text{Re} < 25$), the flow remains predominantly attached, though significant shear layer deformation develops immediately downstream of the constriction.

6.2 Static, Dynamic, and Total Pressure Behavior

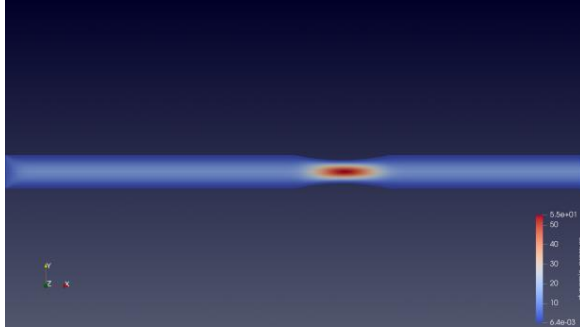
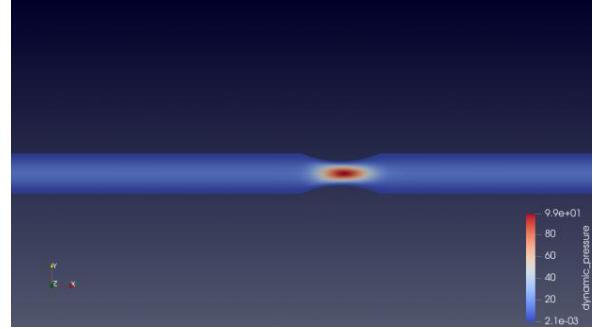
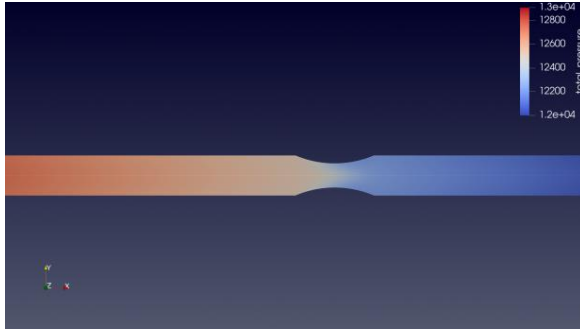
Figure 6 displays the axial distribution and contours of static pressure, dynamic pressure, and total pressure along the vessel.



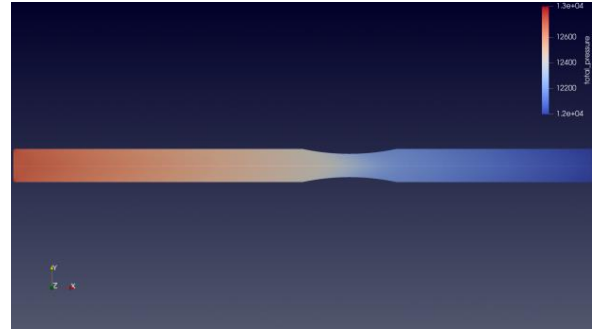
(a) Static pressure: 30% stenosis



(b) Static pressure: 40% stenosis

(c) Dynamic pressure: 30% stenosis ($q_{\max} \approx 55$ Pa)(d) Dynamic pressure: 40% stenosis ($q_{\max} \approx 99$ Pa)

(e) Total pressure: 30% stenosis



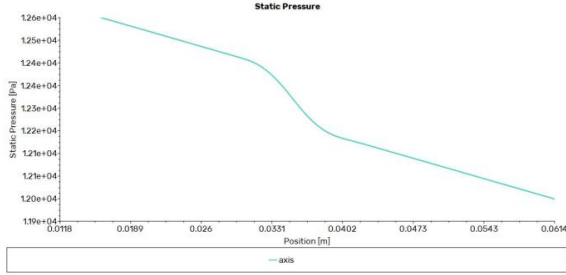
(f) Total pressure: 40% stenosis

Figure 6: Comparison of static, dynamic, and total pressure contours for 30% and 40% stenosis.

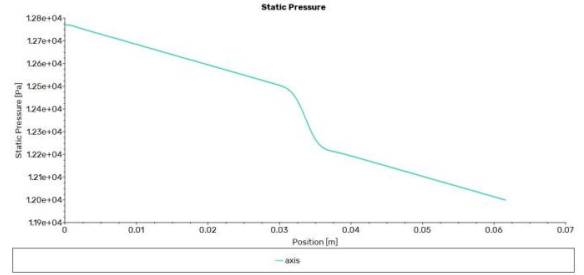
The physical interplay between static and dynamic pressures reveals critical hemodynamic characteristics:

1. **Static Pressure Gradient:** Upstream of the stenosis ($x < 25$ mm), static pressure drops linearly with axial distance, reflecting classical Poiseuille viscous friction. In the converging throat region, a steep drop occurs due to convective fluid acceleration (Bernoulli effect) coupled with intense viscous shearing. Downstream of the throat, viscous dissipation dominates, resulting in poor pressure recovery.
2. **Dynamic Pressure Surge:** Dynamic pressure ($q = \frac{1}{2}\rho u^2$) remains nearly uniform at 3.4 Pa in the straight upstream section, but surges at the stenosis throat, reaching 55 Pa for the 30% case and 99 Pa for the 40% case.
3. **Magnitude Comparison:** Static pressure throughout the artery is of order 10^4 Pa ($\sim 12,000$ Pa), whereas dynamic pressure is of order 10^1 to 10^2 Pa. Consequently, static pressure constitutes over 99% of the total mechanical energy in the arterial system, providing the required wall-normal stress and transmural driving potential.

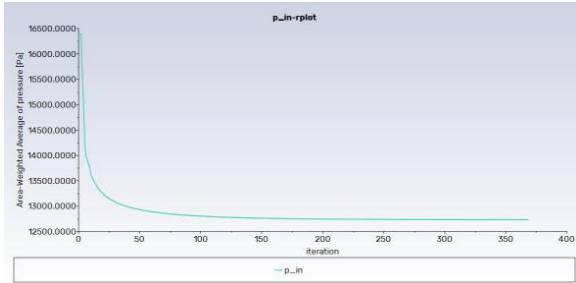
4. **Total Pressure Monotonic Decline:** Because the static pressure drop across the constriction significantly exceeds the dynamic pressure recovery downstream, the total pressure ($p_{\text{tot}} = p_{\text{static}} + q$) decreases monotonically along the vessel axis, illustrating irreversible hydraulic head loss.



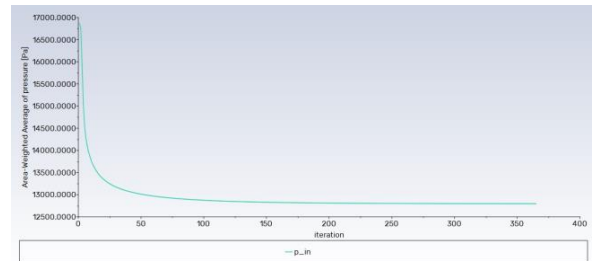
(a) Static pressure along centerline (30% stenosis)



(b) Static pressure along centerline (40% stenosis)



(c) Inlet pressure convergence monitor (30%)



(d) Inlet pressure convergence monitor (40%)

Figure 7: Axial static pressure profiles and iterative convergence monitors of inlet pressure.

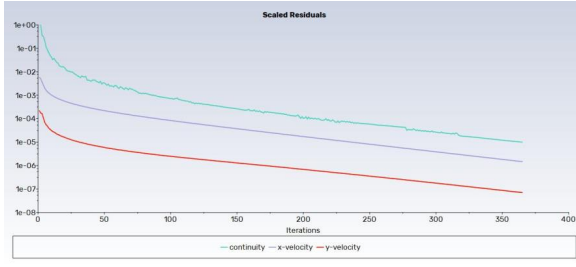
6.3 Hydraulic Resistance and Head Loss Comparison

The quantitative comparison between the two stenosis severities (on the calibrated medium mesh) is summarized in Table 5.

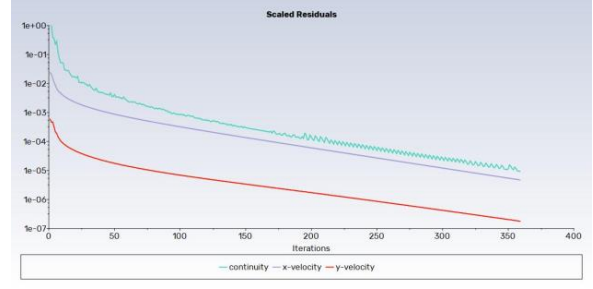
Table 5: Quantitative comparison of hemodynamic parameters between 30% and 40% stenosis.

Occlusion Severity	p_{in} [Pa]	p_{out} [Pa]	Δp [Pa]	ξ
Small (30% Stenosis)	12,733.91	11,999.00	734.91	216.07
Large (40% Stenosis)	12,796.60	11,999.00	797.60	234.47
Difference (Δ)	+62.69 Pa	0.00 Pa	+62.69 Pa	+18.40 (+8.51%)

An increase in stenosis from 30% to 40% causes an 8.51% increase in the pressure loss coefficient. This demonstrates that even moderate increments in plaque thickness compound arterial resistance, forcing the left ventricle to generate elevated pressure to preserve cerebral blood flow.



(a) Residual convergence for 30% stenosis

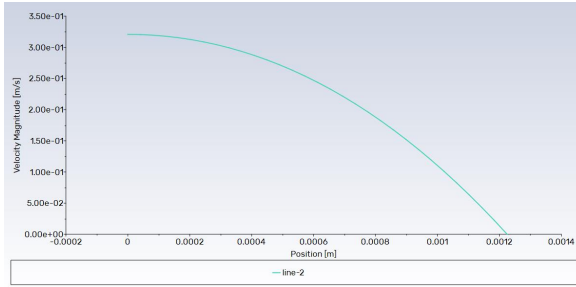
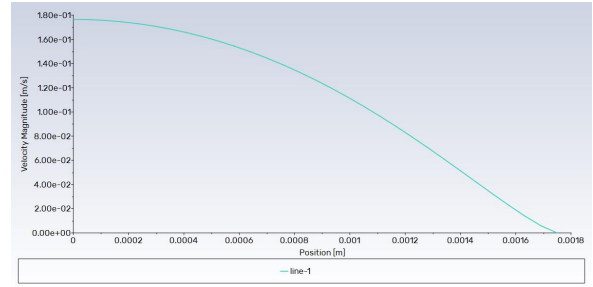


(b) Residual convergence for 40% stenosis

Figure 8: Iterative convergence histories for 30% and 40% stenosis models.

6.4 Cross-Sectional Velocity Profiles at the Stenosis Throat

To evaluate the local flow structure at the narrowest cross-section, radial velocity profiles were extracted across the throat ($x = 30.8$ mm), as plotted in Figure 9.

(a) Radial profile: 30% stenosis throat ($R_{\text{throat}} = 1.225$ mm)(b) Radial profile: 40% stenosis throat ($R_{\text{throat}} = 1.050$ mm)Figure 9: Cross-sectional radial velocity distribution at the stenosis throat from centerline ($r = 0$) to the arterial wall ($u = 0$).

Both profiles display characteristic parabolic velocity distributions, with peak velocities at the centerline ($r = 0$) tapering smoothly to zero at the wall in compliance with the no-slip boundary condition. For the 40% stenosis, the reduced throat radius concentrates the shear rate $\partial u / \partial r$ at the wall, generating substantially elevated wall shear stresses that are clinically known to promote plaque vulnerability and rupture.

7 Parametric Rheological Study: Glycerin as Working Fluid

To investigate the sensitivity of vascular hemodynamics to working fluid rheology, a comparative simulation was performed replacing blood with pure glycerin. The thermophysical properties of glycerin are:

- Density: $\rho_{\text{gly}} = 1259.9 \text{ kg/m}^3$ (an 18.5% increase over blood)
- Dynamic Viscosity: $\mu_{\text{gly}} = 0.799 \text{ Pa} \cdot \text{s}$ (an 18.6-fold increase over blood's $0.043 \text{ Pa} \cdot \text{s}$)

7.1 Grid Independence for Glycerin

A dedicated grid independence verification was conducted for glycerin flow in the 40% stenosis model across coarse, medium, and fine meshes, as presented in Table 6.

Table 6: Grid independence verification for glycerin flow in the 40% stenosed artery.

Grid Level	Elements	p_{in} [Pa]	Δp [Pa]	ξ	Relative Error (%)
Coarse	11,103	27,027.90	15,028.90	3727.2	—
Medium	31,920	26,702.30	14,703.30	3646.5	2.165%
Fine	119,356	27,021.63	15,022.63	3725.7	2.125%

The relative difference between medium and fine grids is 2.12%, validating that the medium grid maintains high numerical precision for highly viscous flows. The corresponding iterative residuals are illustrated in Figure 10.

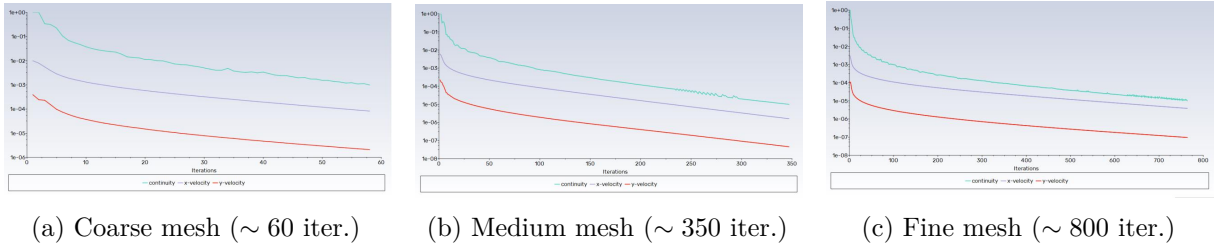
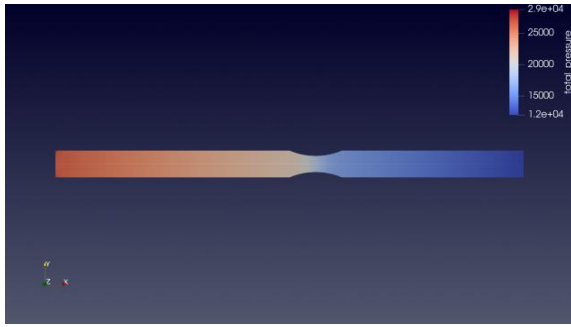
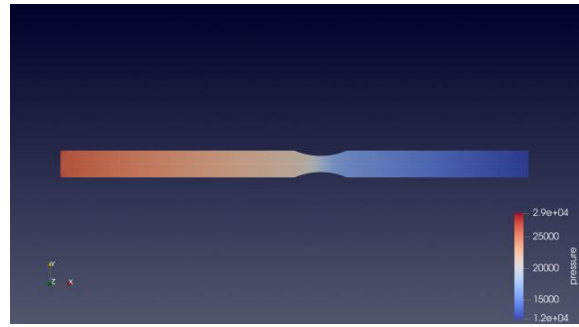
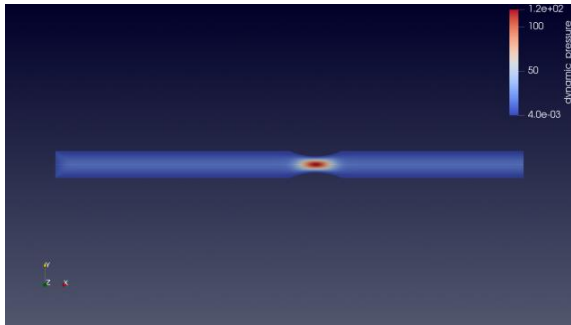
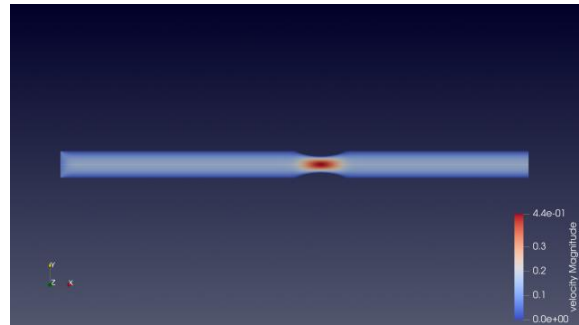
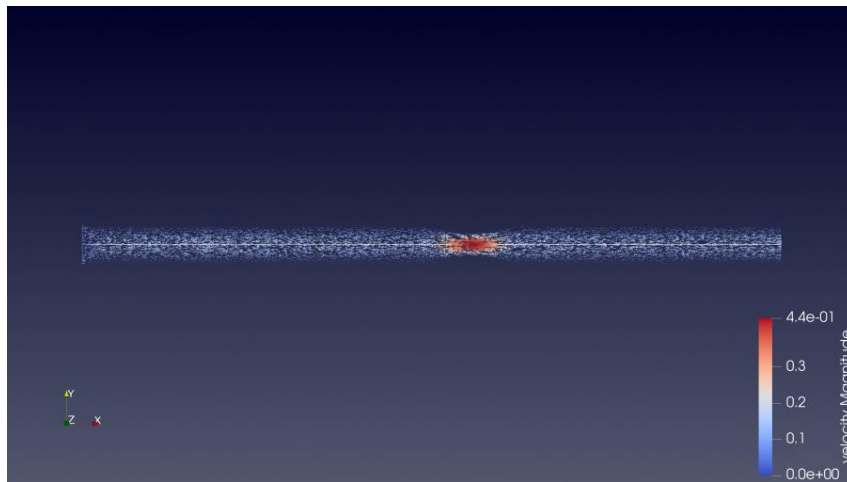


Figure 10: Iterative residual histories for glycerin flow across coarse, medium, and fine grids.

7.2 Flow Contours and Comparative Analysis

Figure 11 depicts the total pressure, static pressure, dynamic pressure, velocity contours, and streamlines for the glycerin simulation.

(a) Total pressure contour ($p_{\max} \approx 29,000$ Pa)(b) Static pressure contour ($p_{\max} \approx 29,000$ Pa)(c) Dynamic pressure contour ($q_{\max} \approx 120$ Pa)(d) Velocity contour ($u_{\max} \approx 0.44$ m/s)

(e) Flow streamlines for glycerin

Figure 11: Hemodynamic and pressure contours for glycerin as the working fluid (40% stenosis).

7.3 Rheological Sensitivity Discussion

Table 7 contrasts the hydrodynamic performance of blood versus glycerin.

Table 7: Comparison of hydrodynamic performance between blood and glycerin in the 40% stenosed artery.

Parameter	Blood	Glycerin	Ratio (Glycerin / Blood)
Density ρ [kg/m ³]	1063.0	1259.9	1.185×
Dynamic Viscosity μ [Pa · s]	0.043	0.799	18.58 ×
Inlet Velocity u_{in} [m/s]	0.08	0.08	1.000×
Inlet Dynamic Pressure q_{in} [Pa]	3.40	4.03	1.185×
Inlet Static Pressure p_{in} [Pa]	12,796.60	26,702.30	2.087×
Static Pressure Drop Δp [Pa]	797.60	14,703.30	18.43 ×
Loss Coefficient ξ	234.47	3646.50	15.55 ×

The results reveal fundamental fluid mechanical principles:

1. **Viscous Dominance:** The total static pressure drop for glycerin ($\Delta p = 14,703.3$ Pa) is 18.43 times higher than that for blood ($\Delta p = 797.6$ Pa). This ratio mirrors the viscosity ratio ($\mu_{gly}/\mu_{blood} = 18.58$) almost exactly. In low-Reynolds laminar flows ($Re \ll 100$), inertial contributions are small compared to viscous shearing forces. Consequently, the momentum equation reduces toward Stokesian/Poiseuille flow, where pressure drop scales linearly with viscosity:

$$\Delta p \propto \mu \cdot u_{in} \quad (8)$$

2. **Loss Coefficient Scaling:** Because the dynamic pressure scaling factor $q_{in} = \frac{1}{2}\rho u_{in}^2$ only scales with density (1.185×), the non-dimensional loss coefficient $\xi = \Delta p/q_{in}$ rises dramatically from 234.47 for blood to 3646.5 for glycerin (a 15.55-fold increase).
3. **Clinical Analogy to Hyperviscosity:** In clinical pathophysiology, elevated blood viscosity (e.g., hypergammaglobulinemia, polycythemia, or elevated fibrinogen) substantially increases trans-stenotic pressure gradients and cardiac workload, demonstrating that hemorheological alterations can severely exacerbate the hemodynamic penalties of arterial stenosis.

8 Conclusions and Engineering Insights

A rigorous computational fluid dynamics investigation of hemodynamic flow through stenosed carotid arteries was conducted. The main findings and engineering conclusions are summarized as follows:

1. **Flow Acceleration and Conservation of Mass:** Constriction of the arterial lumen induces intense local velocity amplification at the stenosis throat (0.32 m/s for 30% stenosis and 0.43 m/s for 40% stenosis, relative to an inlet velocity of 0.08 m/s). The flow remains laminar throughout, with parabolic radial velocity profiles at the throat.
2. **Static vs. Dynamic Pressure Interplay:** Static pressure accounts for over 99% of the total fluid mechanical energy inside the arterial conduit. At the throat, convective acceleration causes a localized surge in dynamic pressure, accompanied by a steep drop in static pressure. Viscous dissipation prevents downstream static pressure recovery, leading to a permanent total pressure loss along the vessel.
3. **Effect of Stenosis Severity:** Increasing the occlusion from 30% to 40% raises the overall trans-stenotic pressure drop from 734.91 Pa to 797.60 Pa, increasing the non-dimensional pressure loss coefficient from $\xi = 216.07$ to $\xi = 234.47$ (+8.51%).
4. **Mesh Verification:** Grid independence was established on three grid levels (coarse, medium, and fine). The relative error between the medium grid (31,920 elements) and the fine grid (119,356 elements) was under 1.9%, confirming spatial convergence.
5. **Rheological Sensitivity:** Substituting blood with glycerin ($\mu = 0.799 \text{ Pa} \cdot \text{s}$) induced an 18.4-fold surge in total pressure drop and inflated the loss coefficient to $\xi = 3646.5$, proving that at low Reynolds numbers, viscous shear dominates over inertial forces in dictating hydraulic resistance.

9 References and Computational Resources

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